

ADVANCEMENTS IN FREQUENCY DOMAIN ANALYSIS OF DUAL CENTERLINE PROPELLANT TANKS FOR SPACE VEHICLE STABILITY AND DESIGN

Han Woong (Brian) Bae ^{*} Jing Pei[†] Justin Ganiban [‡] and Juan Orphee [§]

This paper presents advancements in the frequency domain stability analysis for landing and ascent vehicle configurations with dual centerline propellant tanks, developed in support of Guidance, Navigation, and Control (GNC) insight activities for the Human Landing System project. The study focuses on the impact of axial thrust on the dual tank slosh danger zone solution. The frequency domain stability analysis tool is developed within the Generalized Aerospace Simulation in Simulink® (GLASS) framework at NASA Marshall Space Flight Center. The frequency domain stability analysis tool is used to determine slosh damping and control systems design requirements. The accuracy of the tool's instability predictions is validated against time-domain outcomes within the GLASS simulation environment, showing consistent and anticipated results. Additionally, the method is validated by comparing its results with Frequency Domain Analysis and Comparison Tool Assuming Linearity (FRACTAL), a frequency domain analysis tool with extensive heritage and rigorous verification and validation using NASA Ares I-X and SLS flight data. The findings of this paper contribute to a better understanding of dual-tank slosh instability and support improved design and operation margins of space vehicles.

NOMENCLATURE

δ_{sj}	sloshing mass z-axis displacement in j th tank	m_{sj}	slosh mass in j th tank
ζ_{sj}	slosh damping coefficient in j th tank	m	total vehicle mass
ω_{sj}	slosh natural frequency in j th tank	J	total vehicle m.o.i. about y axis
z	vehicle c.g. z displacement w.r.t. the ref. frame	β_e	thrust vector angle
l_{sj}	sloshing mass location relative to the c.g.	$l_{lim,j}$	aft limit of the j th tank danger zone
ϕ	vehicle pitch angle w.r.t. the ref. frame	l_g	dist. between c.g. and gimbal loc.
k_3	proper axial acceleration	T	total vehicle thrust
U	thrust vector angle expressed in s domain	Θ	vehicle pitch expressed in s domain

INTRODUCTION

Slosh dynamics are a well-known stability concern for space vehicles, impacting both the ascent and descent phases of a mission. The linearized dynamics of a vehicle, as presented in Refs. [1–3],

^{*}Aerospace Engineer, Control Systems Design and Analysis Branch, NASA Marshall Space Flight Center, Huntsville, AL 35808.

[†]Aerospace Engineer, Atmospheric Flight Entry Systems Branch, NASA Langley Research Center, Hampton, Virginia, 23681.

[‡]Aerospace Engineer, Control Systems Design and Analysis Branch/ESSCA, NASA Marshall Space Flight Center, Huntsville, AL 35808.

[§]Aerospace Engineer, Control Systems Design and Analysis Branch, NASA Marshall Space Flight Center, Huntsville, AL 35808.

typically model slosh mode as either a spring-mass or a pendulum. The mechanical model parameters representing slosh mass, frequency, etc., are often scheduled based on flight conditions, such as fill level. During the Saturn Era, a rule-of-thumb known as the slosh "danger zone" was established, which provided an indication of the conditions for likely instability or control challenges ([4]). In a space vehicle with a single centerline slosh tank under thrust vector control, the "danger zone" occurs when the slosh mass location is between the vehicle center of gravity (CG) and the center of percussion, also called the instantaneous center of rotation.

If inside the danger zone, the slosh mass has unfavorable phasing with respect to rigid body control, but not necessarily phase unstable. As long as the slosh curve does not go counterclockwise about the stability critical point (0 dB, -180 degree) the closed-loop system is stable. When the slosh is phase unstable such that slosh curve wraps about the stability critical point (0 dB, -180 degree), the control effector actions indirectly add energy to the slosh mode when attempting to actively control rigid body attitude. Figure 1 provides a notional Nichols plot with an unfavorably phased, but stable propellant slosh mode with violations in the disc margin. Slosh baffles and filter design are typically required to meet margin requirements.

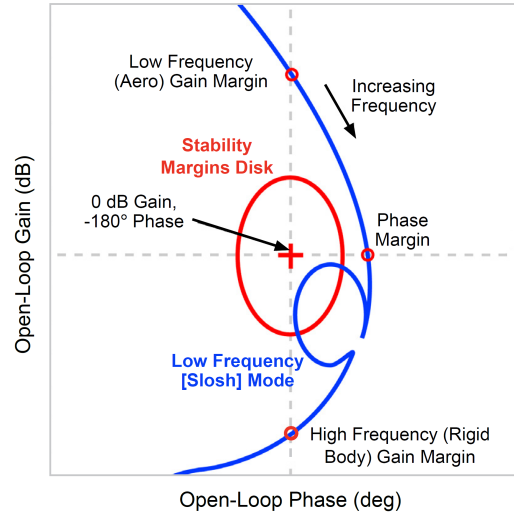


Figure 1: Notional Nichols Plot with Adversely Phased Propellant Slosh Model [5]

Recent advances ([6]) showed that the aft end of the danger zone extends beyond the center of gravity and is determined by the vehicle axial acceleration, k_3 (consistent with the nomenclature used in the original literature [1]). This fact was subsequently verified in Ref. [7]. The slosh danger zone for j th tank with axial proper acceleration effect is expressed as:

$$\frac{k_3(m - m_{sj})}{m\omega_{sj}} < l_{sj} < -\frac{J}{ml_g} \quad (1)$$

where the term $-\frac{J}{ml_g}$ is a center of percussion, Q , with reference to the center of gravity. The term $\frac{k_3(m - m_{sj})}{m\omega_{sj}}$ is the aft limit of the danger zone, measured with reference to the center of gravity. The locations of the center of percussion, slosh masses, and aft limits are delineated in Fig. 2.

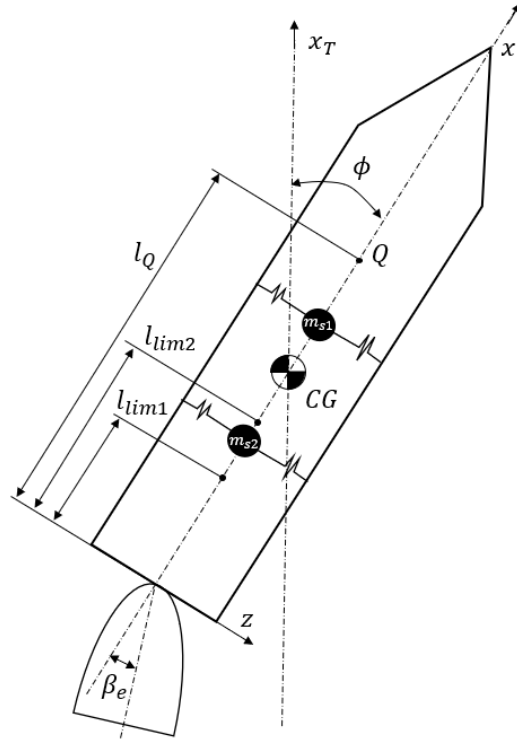


Figure 2: Simplified space vehicle dynamics with multiple centerline slosh masses and gimbaled thrust vector.

Furthermore in Ref. [7], the concept of the slosh danger zone was extended to a space vehicle with a two-tank configuration, stacked along the centerline. This was an improvement to the classic single tank originally presented in Ref. [4] consider liquid propelled vehicles generally consist of two tanks, fuel and oxidizer. The dual tank formulation accounts for the interaction effects between the tanks. In the case where the frequency separation between the slosh modes is large and/or mass fraction ratios are small, the danger zones/stability boundaries for each mode closely resemble the single-tank solution. However, in the limiting case where the slosh modes are nearly identical in mass and frequency, the stability boundaries for both tanks deviate significantly from the single-tank solution. Figure 3 illustrates an example in which the slosh modes are individually phase stable when considering the single-tank solution, but in dual tank solution one of the sloshes becomes unfavorably phased and unstable. The example is based on the dynamics and state space model formulation described in the later part of the paper, with slosh modal frequencies closely aligning with the natural frequency of the control system. Unfortunately, due to the drastic increase in the complexity of the transfer function that describes the attitude dynamics, obtaining a closed-form solution to the danger zone was no longer tractable as demonstrated by Jing et. al. [7]. Hence, numerical analysis is required to provide insight into the impact the secondary slosh mass has on the stability boundary of the primary slosh mass and vice versa. In a recent assessment by the NASA Engineering Safety Center (NESC), the dual tank analysis proved to be valuable in understanding the fundamental physics of the launch vehicle problem providing justifications for flying with reduced stability margins ([8], [9]). Reference [5] provides a summary of that assessment

along with recommended best practices for the treatment of flight control stability margins in the presence of slosh dynamics.

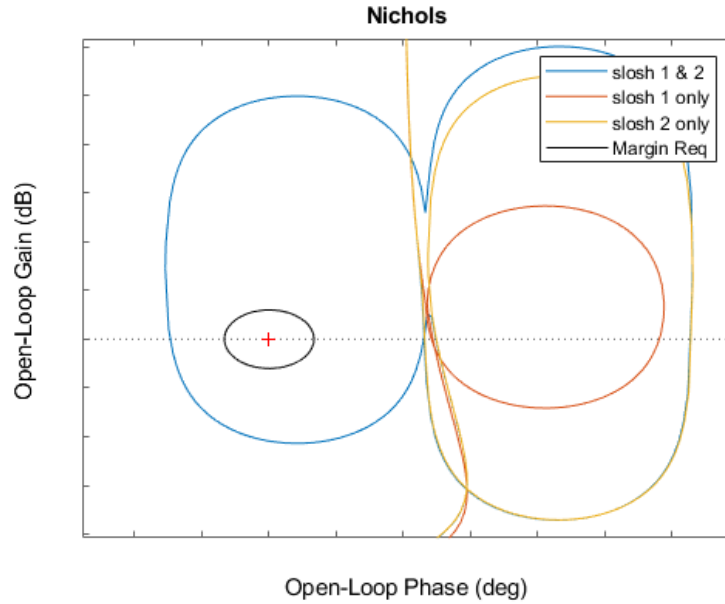


Figure 3: Nichols charts illustration of single tank vs. dual tank configurations.

Figure 5 illustrates the corresponding danger zone plot for the dual tank configuration considered in Nichols plot shown in Fig. 3. The x-axis represents all possible axial locations for slosh mass #1, and the y-axis represents all possible axial locations for slosh mass #2. The black "x" denotes the current slosh mass location. The green dashed line delineates the single tank danger zone boundary for slosh mass #1, and the same goes for the red dashed line to slosh mass #2. The yellow region indicates both slosh masses being favorably phased, the orange region indicates only slosh mass #2 being unfavorably phased, the blue region indicates only slosh mass #1 being unfavorably phased, and the navy blue region indicates both slosh masses being unfavorably phased. Figure 4 shows the descriptions of the color coding associated with the dual tank danger zone plot. In a single tank configuration for each slosh mass, the slosh is expected to be favorably phased and stable given the current slosh location falls outside the danger zone boundaries of the single tank configuration. However, in the dual tank configuration, the slosh mass is situated in the orange region, indicating an unfavorably phased slosh mass #2. The Nichols result in Fig. 3 is consistent with the danger zone outcomes depicted in Fig. 5. For example, Fig. 3 shows that sloshes ("slosh 1 only" and "slosh 2 only") for both tanks are favorably phased such that the slosh is stable if modeled in single tank configurations, but in dual tank configuration ("slosh 1 & 2") one of the sloshes becomes unfavorably phased and unstable.

The dual tank danger zone formulation discussed in Ref. [7] did not account for the effects of axial acceleration, k_3 . However, during Earth ascent, a launch vehicle can experience axial acceleration beyond 3 g's, and the k_3 term could have an impact on the dual tank danger zone solution, hence the motivation for the present work.

Furthermore, the dual tank danger zone formulation discussed in Ref. [7] neglects the inclusion

	Slosh mass #1 and #2 favorably phased.
	Slosh mass #2 unfavorably phased.
	Slosh mass #1 unfavorably phased.
	Slosh mass #1 and #2 unfavorably phased.

Figure 4: Dual tank danger zone color descriptions.

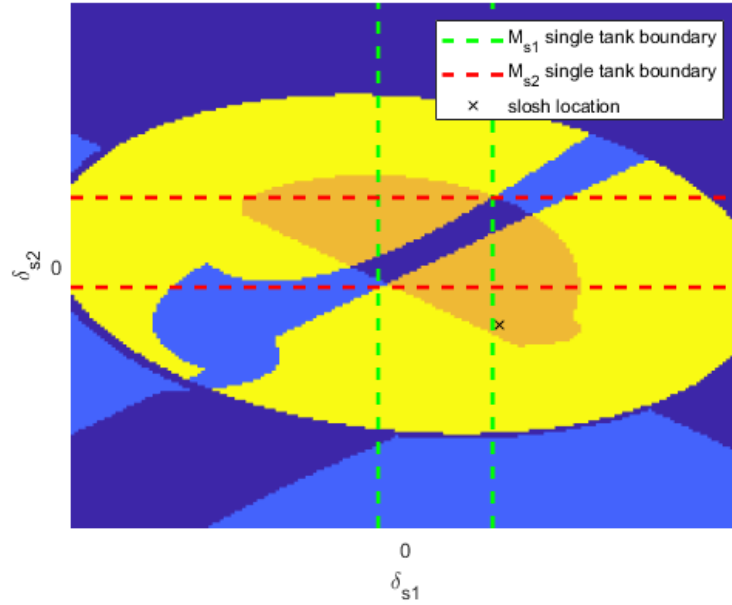
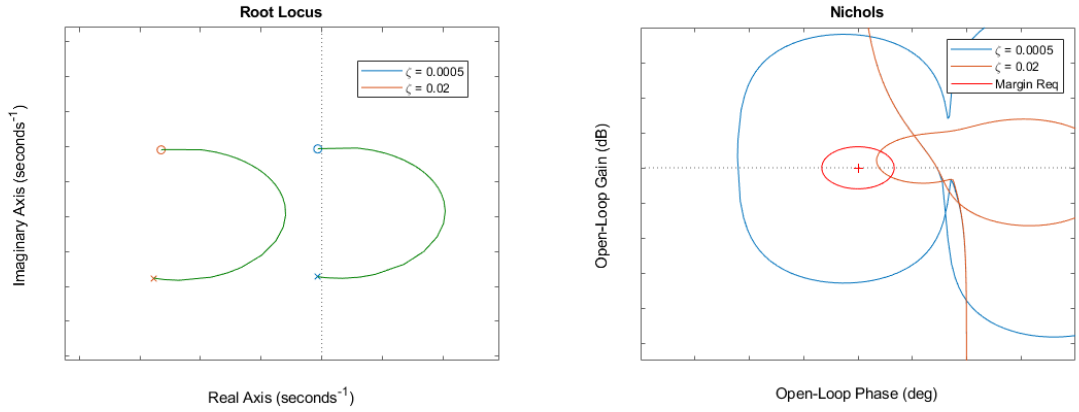


Figure 5: The example of dual tank danger zone plot.

of the slosh damping. The impact of the slosh damping is emphasized in [6]. This paper incorporates an examination of the slosh damping effect and provides evidence of its sensitivity. As mentioned previously in this paper, the term "danger zone" in paper is not designated as a zone denoting unstable slosh, but characterized as a zone that has the slosh Nichols curve that is phased unfavorably, and trends toward the stability critical point (-180 deg and 0 dB). This definition stems from the realization that while damping can be employed for slosh stabilization, it often falls short of meeting stability margin requirements in cases involving tight control design. Figure 6b shows a Nichols example where slosh exhibits instability under minimal bare wall damping with a damping coefficient value of 0.05% and even with a high damping coefficient, 2%, the stability margin is breached. Likewise, Fig. 6a shows the 2% damping root locus curve not crossing the right-hand side plane, the stability margin is still violated as seen in the Nichols plot. Thus, the danger zone is defined as a region of unfavorably phased slosh with consideration of damping in the formulation of the danger zone in this paper.

Lastly, the slosh parameterization has been modified to a mass-spring-damper configuration from the pendulum formulation considered in Ref. [7]. This adjustment was made in response to the persistent confusion and discrepancies found in the literature regarding the accurate assumptions



(a) Damping sensitivity shown in Root Locus Curves (b) Damping sensitivity shown in Nichols Curves

Figure 6: Damping sensitivity in Root Locus and Nichols.

related to the sloshing mass locations. In Refs. [10] and [11], the authors suggest locating the spring-mass mass at the same elevation as the pendulum mass and not at the pendulum hinge location as depicted in Fig. 7a. On the other hand, Refs. [6] and [7] assume the pendulum hinge point is aligned with the spring-mass mass as depicted in the analogy in Fig. 7b. While this assumption maintains dynamic consistency in terms of forces and torques induced by the moving mass, employing this model with slosh model parameters obtained with methods in Refs. [10] and [11] can introduce modeling errors. This is due to the dependence of the stationary slosh mass and inertia on the locations of the moving slosh mass. Furthermore, Ref. [7] used the pendulum hinge point location, and not the pendulum mass location, as a slosh reference point to determine its stability with the slosh danger zone. If the case in Fig. 7a is used, then the danger zone reference point may deviate by the length of the pendulum arm. If the case in Fig. 7b is used is chosen, it contradicts the slosh parameter determination requirement outlined in Refs. [10] and [11]. To avoid these discrepancies, the danger zone formulation adopts the more conventionally used spring-mass model.

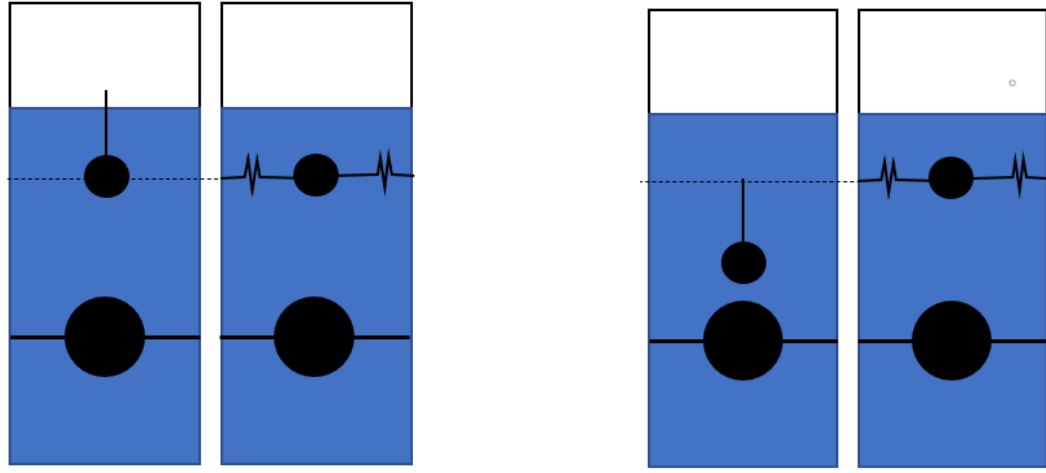
In this work, the modified dual tank danger zone formulation was validated with the Frequency Domain Analysis and Comparison Tool Assuming Linearity (FRACTAL), a well-established frequency domain analysis tool with an extensive heritage, having undergone rigorous verification and validation processes leveraging NASA Ares I-X and SLS flight data. Furthermore, additional checks were performed in the time domain in the GLASS simulation environment showing expected outcomes.

FORMULATION

The formulation of the dynamical equations of motion used for the dual tank danger zone study is stemmed from [1] and [3].

Dynamics

The dynamics involving rigid body and slosh are derived from the linearized planar equations of motions as presented in [1]. To minimize the complexity, the nozzle dynamic, flex dynamic, and aerodynamic were not taken into consideration in the formulation.



(a) Pendulum mass and mass-spring mass located on the same height

(b) Pendulum hinge point and mass-spring mass located on the same height

Figure 7: Two different cases of pendulum and mass-spring equivalency analogies

$$\ddot{\delta}_{sj} + 2\zeta_{sj}\omega_{sj}\dot{\delta}_{sj} + \omega_{sj}^2\delta_{sj} = -\ddot{z} - l_{sj}\ddot{\phi} + k_3\phi, \quad j = 1 \dots n \quad (2)$$

$$\ddot{z} = F_r + F_s \quad (3)$$

$$\ddot{\phi} = M_r + M_s \quad (4)$$

$$F_r = T\beta \quad (5)$$

$$M_r = -l_g T\beta \quad (6)$$

$$F_s = -\sum_{j=1}^n m_{sj}\ddot{\delta}_{sj} \quad (7)$$

$$M_s = -\sum_{j=1}^n m_{sj}l_{sj}\ddot{\delta}_{sj} - k_3\sum_{j=1}^n m_{sj}\delta_{sj} \quad (8)$$

State Space Model

The rigid body and slosh equations shown in Eqs. 2 to 8 are subsequently placed in the state-space format similar to Ref. [3] such the dynamic coupling terms become more apparent. Two slosh masses are considered ($j = 2$).

$$E\dot{x} = Ax + Bu, \quad E = \begin{bmatrix} E_r \\ E_s \end{bmatrix}, \quad A = \begin{bmatrix} A_r & A_{r/s} \\ A_{s/r} & A_s \end{bmatrix} \quad (9)$$

$$y = Cx + Du, \quad C = [1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad D = 0 \quad (10)$$

$$x = [\phi \quad \dot{\phi} \quad \dot{z} \quad \delta_{s1} \quad \dot{\delta}_{s1} \quad \delta_{s2} \quad \dot{\delta}_{s2}]^T \quad (11)$$

$$A_r = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ mk_3 & 0 & 0 \end{bmatrix} \quad (12)$$

$$A_{r/s} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -m_{s1}k_3 & 0 & -m_{s2}k_3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (13)$$

$$A_{s/r} = \begin{bmatrix} 0 & 0 & 0 \\ k_3 & 0 & 0 \\ 0 & 0 & 0 \\ k_3 & 0 & 0 \end{bmatrix} \quad (14)$$

$$A_s = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\omega_{s1}^2 & -2\zeta_{s1}\omega_{s1} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\omega_{s2}^2 & -2\zeta_{s2}\omega_{s2} \end{bmatrix} \quad (15)$$

$$B = \begin{bmatrix} 0 \\ -l_g T \\ T \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (16)$$

$$E_r = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & J & 0 & 0 & m_{s1}l_{s1} & 0 & m_{s2}l_{s2} \\ 0 & 0 & m & 0 & m_{s1} & 0 & m_{s2} \end{bmatrix} \quad (17)$$

$$E_s = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & l_{s1} & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & l_{s2} & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (18)$$

The transfer function from thrust vector angle U to vehicle pitch angle Θ is obtained by

$$\frac{\Theta}{U} = C(sI - E^{-1}A)^{-1}E^{-1}B \quad (19)$$

This transfer function is then used to find the poles and zeros of each slosh. When the natural frequency Ω of the pole is greater than the zero ($\Omega_p^2 > \Omega_z^2$) the angle of departure leans toward the left half of the s-plane, which is favorably phased. The angle of the departure is the angle at which the root locus originates from the open-loop pole. On the other hand, when the slosh pole is less than the zero ($\Omega_p^2 < \Omega_z^2$), the angle of departure leans toward the right half of the s-plane; thus, leading to an unfavorably phased scenario with an increased risk of slosh instability. The example of unfavorably phased root locus curves is seen in Fig. 6a.

RESULTS

This section delves into the verification of the dual tank danger zone solution in both frequency and time domains using the FRACTAL and GLASS tools. Furthermore, the proper axial acceleration and slosh damping sensitivities were explored.

FRACTAL Verification

As a verification step, Nichols plots from the open loop transfer function $\frac{\Theta}{U}$ were compared the reduced order model presented in the previous section with FRACTAL. FRACTAL is a full 3-axis stability analysis tool while the dual tank danger zone solution assumes planar dynamics. Figure 8 considers two scenarios: a) both slosh modes are stable, and b) one unstable slosh mode. In both cases, frequency response of the reduced order model closely resembles that of FRACTAL. Note that the reduced order model open loop transfer function used in the Nichols plot was multiplied with the thrust vector control (TVC) actuator model.

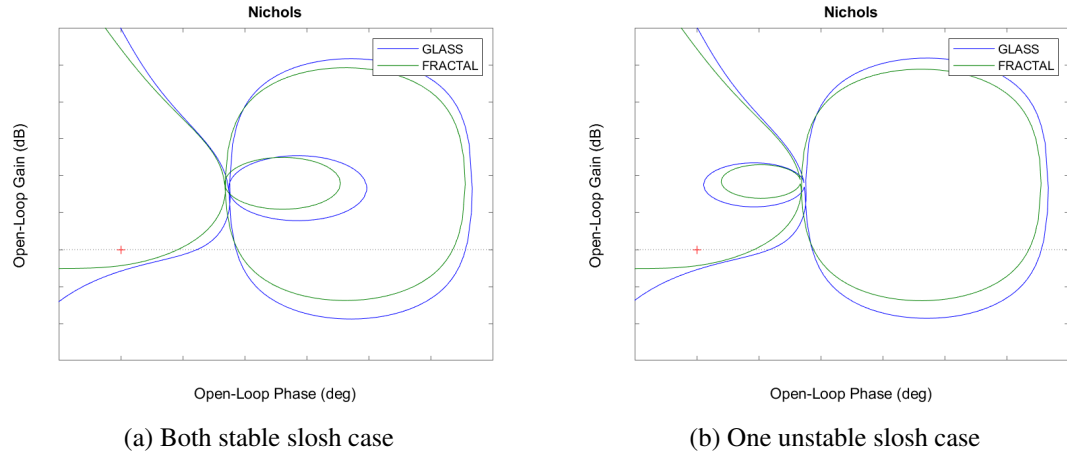


Figure 8: GLASS and FRACTAL comparison

k_3 Sensitivity

The impact of the proper axial acceleration on the widening of the slosh danger zone was first discussed in Ref. [6] for the single tank configuration. With the inclusion of the k_3 term, it is observed that the single tank danger zones are widened as the lower limits creep below the center of mass (0 marker) in Fig. 9. Likewise considering the dual tank danger zone solution, the orange

region, which leads to the unfavorably phased slosh #1, grew larger. Whereas, when the k_3 term is excluded, the x marker resides in the yellow region, indicating that both slosh masses are favorably phased.

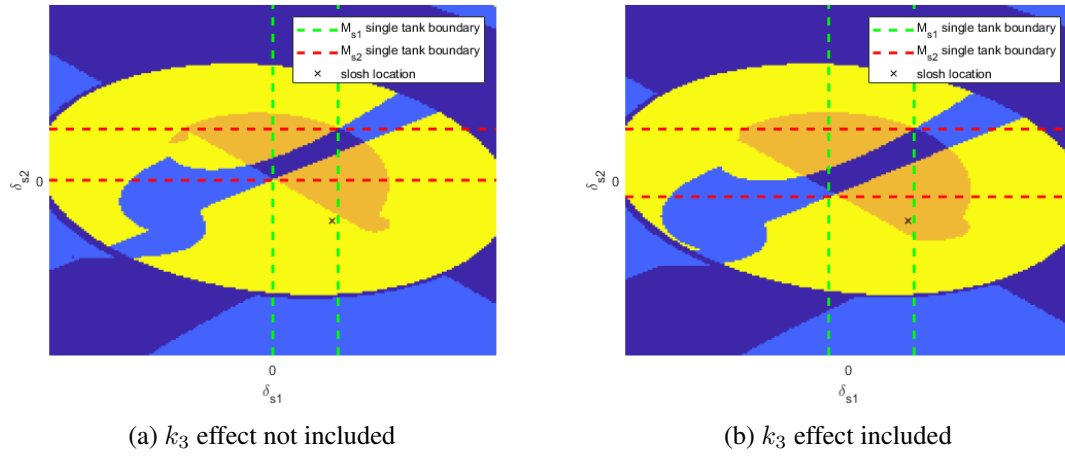


Figure 9: Sensitivity of k_3 effect.

Slosh Damping Sensitivity

The effect of damping on the danger zone is explored, revealing significant deviations from the baseline (where both tanks have $\zeta = 0.05\%$) as ζ values deviate between the tanks. Figure 10 demonstrates a considerable shift in the danger zone, causing a shift from unfavorably phased slosh # 1 to unfavorably phased slosh # 2 while the slosh mass locations remain fixed.

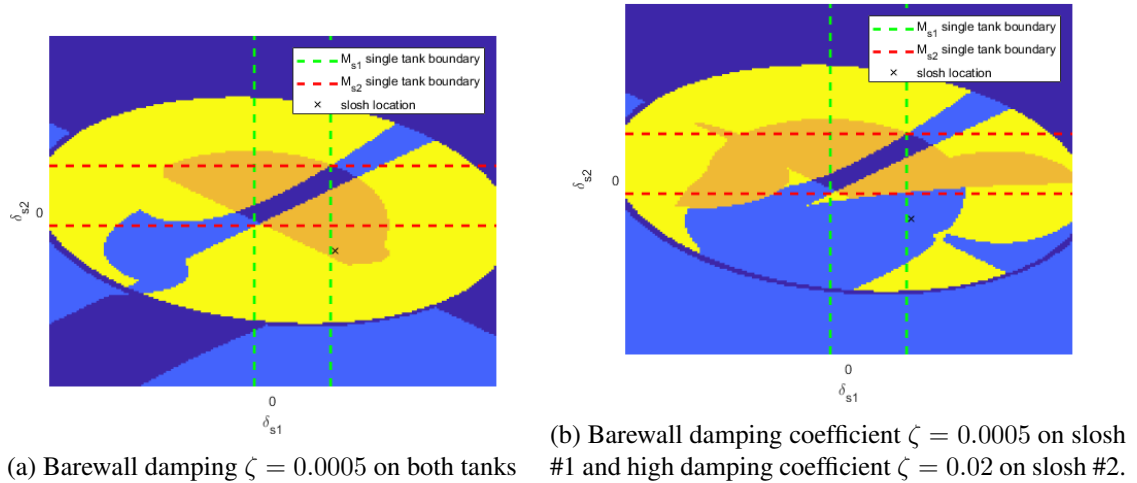


Figure 10: Sensitivity of the slosh damping coefficient.

GLASS 6 DOF verification

To verify the validity of the modified dual tank danger zone solution, the GLASS 6 DOF simulation results were meticulously examined. The scenario involves the braking burn phase of the lunar lander configuration with two centerline propellant tanks. A danger zone plots at T350 (with T250 representing the start of the braking burn) is depicted in Fig. 9. The commanded attitude is held constant in the simulation to ensure cleaner verification of the results. Furthermore, the simulation assumes a constant damping coefficient that is not a function of wave height. The primary focus of the verification is to perform a time domain verification of the k_3 sensitivities shown in Fig. 9 where for the prescribed locations of slosh masses, slosh mass #2 becomes unfavorably phased when the k_3 term is included.

Table 1: Slosh Unfavorably Phased Stability Expected by Configurations and Axial Acceleration Assumptions

Danger Zone Metric	Axial Acc.	T250	T300	T350	T400	T450
Single Tank	k_3 excluded	None	None	Slosh #2	Slosh #2	Slosh #2
	k_3 included	None	None	Slosh #2	Slosh #2	Slosh #2
Dual Tank	k_3 excluded	None	None	None	None	None
	k_3 included	Slosh #2	Slosh #2	Slosh #2	Slosh #2	Slosh #2

The prediction by the dual tank danger zone analysis with the k_3 term included aligns with the GLASS/FRACTAL frequency domain results. It is also corroborated by the time domain results from the GLASS 6 DOF simulation. Large and unstable growth in the slosh mass #2 velocity is observed in Fig. 11 when k_3 term is included at various damping levels. The damping values are the same for both tanks in this particular test. Conversely, when the k_3 term is disregarded, the simulation shows stable slosh with a very small limit cycle oscillation.

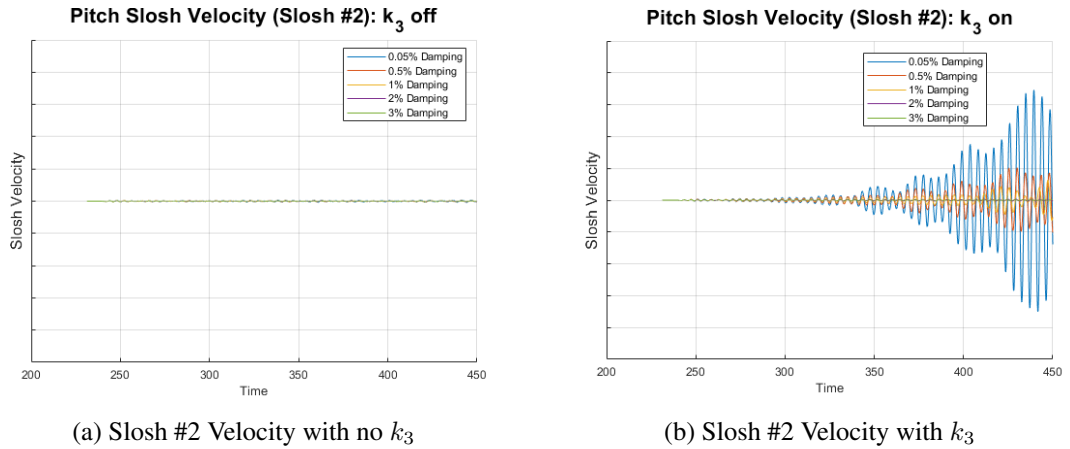


Figure 11: Slosh Velocity in z body axis with k_3 term included and excluded.

Also, as predicted by the Nichols chart using the transfer function $\frac{\Theta}{U}$ with varying slosh damping values, the 6 DOF results exhibit unstable slosh behavior when the damping is below 1%, marginal stability when the damping is at 1% and stable behavior for damping values exceeding 1%

as depicted in Fig. 12.

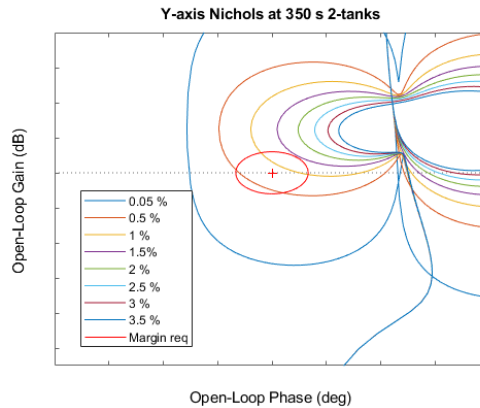


Figure 12: Nichols charts with varying slosh damping values.

CONCLUSION

In this paper, advancements in the frequency domain stability analysis of landing and ascent vehicle configurations with dual centerline propellant tanks have been presented. The study focused on the impact of axial thrust on the dual tank slosh danger zone solution. Furthermore, this paper presented the formulation and validation of the dual tank danger zone analysis tool, developed within the Generalized Aerospace Simulation in Simulink® (GLASS) framework.

The accuracy of the dual tank danger zone analysis tool predictions was validated against time-domain outcomes within the GLASS simulation environment, showing consistent and anticipated results. Additionally, the method was validated by comparing its results with FRACTAL in the frequency domain.

The findings of this paper contribute to a better understanding of dual-tank slosh instability and support improved design and operation margins of space vehicles. Sensitivity analyses were conducted to explore the impact of proper axial acceleration (k_3) and slosh damping coefficients on the dual tank danger zone results. The study revealed that the inclusion of the k_3 term widens the dual tank danger zone, and different damping coefficient values for each tank can significantly affect the dual tank danger zone results.

Overall, the proposed frequency domain stability analysis tool provides valuable insights into the dynamics of dual centerline propellant tanks, offering a useful method for engineers and designers involved in the stability and control aspects of space vehicles during either planetary ascent or descent phases.

ACKNOWLEDGMENT

The authors acknowledge Nick Olson for his contributions to the development of GLASS and standalone slosh models, Victor Tapia for his work on thrust vector control (TVC) control design, Ravi Purandare for providing slosh models and insights, John Wall for his advice and review, and the Human Lander System (HLS) Guidance Navigation and Control (GNC) team and Marshall Space Flight Center Control Systems Design and Analysis Group for their valuable support and review.

REFERENCES

- [1] J. A. Frosch and D. P. Vallely, "Saturn AS-501/S-IC Flight Control System Design," *Journal of Spacecraft*, Vol. 4, No. 8, 1967, pp. 1003–1009.
- [2] A. L. Greensite, "Analysis and Design of Space Vehicle Flight Control Systems, Volume VII-Attitude Control During Launch," *Journal of Spacecraft*, Vol. NASS CR-826, 1967, pp. 41–46.
- [3] J. Orr, "State Space Implementation of Linear Perturbation Dynamic Equations for Flexible Launch Vehicles," *AIAA Guidance, Navigation, and Control Conference*, No. 2009–5962, 1967.
- [4] H. F. Bauer, "Stability Boundaries of Liquid-Propelled Space Vehicles with Sloshing," *Journal of Spacecraft*, Vol. 1, No. 7, 1967, pp. 1583–1589.
- [5] T. VanZwieten, J. Wall, C. Dennehy, D. Dyer, R. Hall, W. Benson, and J. Pei, "Consideration Regarding the Treatment of Launch Vehicle Flight Control Stability Margin Reductions with Emphasis on Slosh Dynamics," AAS 23-151.
- [6] J. A. Ottander, R. A. Hall, and J. F. Powers, "Practical Methodology for the Inclusion of Nonlinear Slosh Damping in the Stability Analysis of Liquid-propelled Space Vehicles," *AIAA SciTechForum*, No. 2018-2097, 2018.
- [7] J. Pei, "Analytical Investigation of Propellant Slosh Stability Boundary on a Space Vehicle," *Journal of Spacecraft and Rockets*, Vol. 58, No. 5, 2021.
- [8] "Launch Vehicle Flight Control Stability Margin Reduction Considerations," 2022. NESC Technical Bulletin No. 22-05.
- [9] "Treatment of Slosh Stability Margin Reductions for Human-Rated Launch Vehicles," 2022. NESC Technical Bulletin No. 22-06.
- [10] F. T. Dodge, *The New Dynamic Behavior of Liquids in Moving Containers*. San Antonio, Texas: Southwest Research Institute, 2000.
- [11] N. Aeronautics and S. Administration, *Propellant Slosh Loads*. Hampton, Virginia: National Aeronautics and Space Administration, 1968.